

INDEPENDENT ANALYTICAL CASE STUDY

Reducing Demurrage Cost Exposure in Saudi Arabian Port Clearance

A Process & Documentation-Risk Analysis of HS Code Classification in Saudi Import Clearance

Prepared by

Asad Ullah

Supply Chain & Logistics Professional | Warehouse Operations | Inventory Management
asadullah3402@gmail.com | linkedin.com/in/asadullah-logistics | Prayagraj, India

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Note: This is an independent research and process-analysis exercise undertaken to build subject-matter understanding of Saudi Arabian trade compliance. It does not represent prior employment or client engagement. All figures are drawn from publicly available industry sources, regulatory guidance, and reported case examples, cited in the References section.

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1. Introduction

Saudi Arabia's logistics and supply chain sector sits at the center of the Kingdom's Vision 2030 economic diversification programme. As the country invests heavily in ports, free zones, and digital customs infrastructure, the efficiency of import clearance has become a direct driver of national competitiveness — and a direct cost line for every business that moves goods through Jeddah Islamic Port, King Abdulaziz Port (Dammam), and the Kingdom's other entry points.

This case study was undertaken as an independent analytical exercise to understand one specific, recurring friction point in that system: the relationship between documentation accuracy — particularly Harmonized System (HS) code classification — and the cost, time, and profit consequences that follow when that documentation is wrong. The objective is not to describe a hypothetical inefficiency, but to map a real, well-documented process, identify precisely where errors originate, and propose a grounded, low-cost intervention.

The motivation for this analysis is professional rather than academic: as a candidate targeting Logistics Coordinator and Supply Chain roles in the Saudi market, understanding how documentation failures propagate into demurrage, fines, and strained business relationships is directly relevant to the day-to-day responsibilities of the role — and to the kind of process-discipline thinking that Gulf employers in FMCG, warehousing, and trade logistics consistently look for in entry-level hires.

2. Background

2.1 The Saudi Customs Environment

Import clearance in Saudi Arabia is administered by the Zakat, Tax and Customs Authority (ZATCA), a Saudi government body combining the country's tax and customs functions. Commercial declarations are filed through FASAH (Facilitate Sea, Air, and Highway Customs Clearance), the Kingdom's single-window digital platform connecting importers, customs brokers, shipping lines, and regulatory agencies such as the Saudi Food and Drug Authority (SFDA) and the Saudi Standards, Metrology and Quality Organization (SASO).

Most commercial importers do not file declarations themselves. Instead, they engage a ZATCA-accredited customs broker (مُخَلِّص, Mukhallas), who prepares the import declaration (Bayan), selects the applicable HS code, calculates duty, and coordinates any additional regulatory approvals the product category requires.

2.2 Why HS Codes Matter

The Harmonized System is a globally standardized numerical classification maintained by the World Customs Organization and used by over 200 countries. Every traded product is assigned a 6–10 digit code based on its material composition, function, and degree of production — the code determines the duty rate applied, the regulatory approvals required, and whether the shipment is treated as routine or restricted. A container of black pepper and a container of crude oil are governed by entirely different codes, different duty schedules, and different inspecting authorities.

2.3 Trusted-Trader Fast Tracking

ZATCA operates an Authorized Economic Operator (AEO) programme, under which importers with a strong compliance history receive priority “green channel” clearance — reducing, but not eliminating, the likelihood of physical inspection. This mirrors similar programmes internationally, such as the United States' Customs Trade Partnership Against Terrorism (C-TPAT).

3. Problem Statement

Despite Saudi Arabia's significant modernization of its customs infrastructure over the past decade, documentation errors — particularly HS code misclassification — remain one of the leading causes of port delay, demurrage cost, and financial penalty for importers. Saudi ports allow a limited free-time window (industry guidance points to roughly a 3-day Bayan filing expectation, with carriers typically allowing 5–7 days before container detention charges begin) before charges accrue. Once that window expires, demurrage is billed on a per-container, per-day basis, and can escalate quickly.

Research question: Where, structurally, do documentation errors originate in the Saudi import clearance process, why do they recur despite modernized digital systems, and what is the most realistic, low-cost intervention available to an entry-level supply chain professional's sphere of influence?

This is not a marginal issue. Global benchmarking suggests that a substantial share of customs declarations — across markets, not Saudi Arabia alone — contain some form of documentation error, and that HS misclassification is consistently cited as a leading cause. Applied to Saudi Arabia's import volumes, even a fraction of that error rate represents a measurable, recurring cost to importers across FMCG, industrial, and consumer goods sectors.

4. Methodology

As an independent desk-based analysis (not a fieldwork or client engagement study), this case study draws on secondary research — ZATCA and FASAH public guidance, licensed customs-broker and freight-forwarder publications, international trade compliance literature (including a comparative review of the United States' CBP model), and specific reported case examples of Saudi clearance delays. The approach combined three methods:

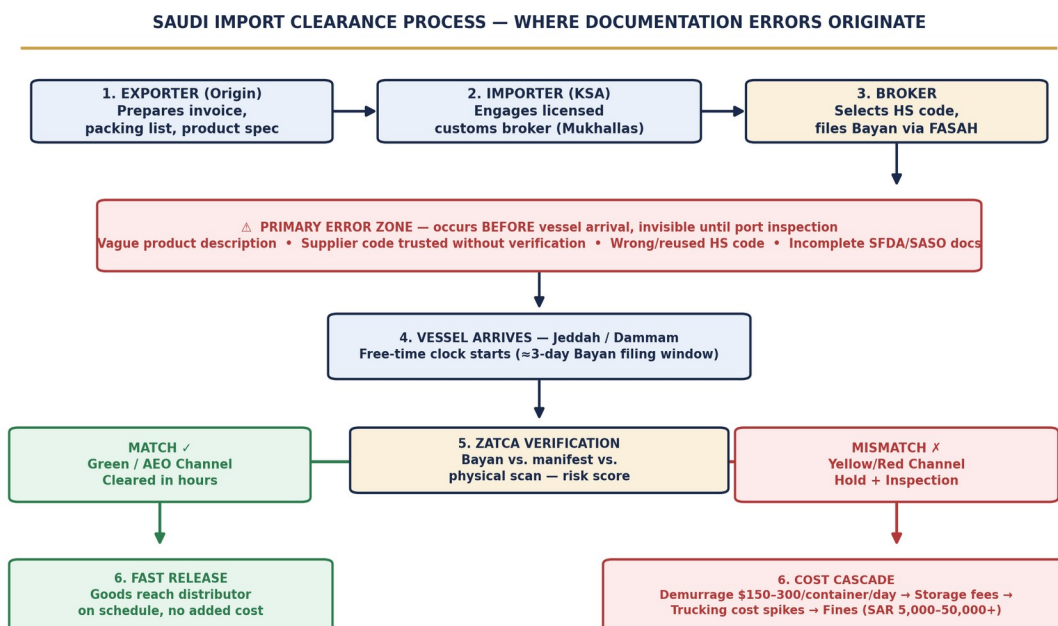
- Process mapping — reconstructing the step-by-step Saudi import clearance sequence, from exporter documentation through vessel arrival, ZATCA verification, and channel assignment, to identify the exact point(s) at which an error can enter the system undetected.
- Comparative regulatory analysis — benchmarking Saudi Arabia's importer/broker liability structure against the United States' “reasonable care” standard (19 U.S.C. § 1484) to identify structural differences in how accountability is assigned and enforced.
- Case triangulation — cross-referencing three independently reported Saudi clearance incidents (chemical/solvent, textile, and electronics import categories) to test whether the process-mapping findings held across different product types and failure modes.

Limitations: this analysis relies on publicly reported figures and industry-published case examples rather than primary ZATCA enforcement data, which is not publicly disclosed at the shipment level. Cost figures should therefore be read as representative industry benchmarks rather than official statistics.

5. Presentation of Findings

5.1 Process Flow and the Primary Error Zone

The process map below traces a shipment from origin documentation through to final release or cost cascade. The critical finding is that the point of error — an imprecise product description, an unverified supplier-provided code, or an incomplete regulatory certificate — is created before the vessel ever leaves for Saudi Arabia, yet it remains invisible to any party until ZATCA's verification step, days or weeks later.



Source: Author's process mapping, synthesized from ZATCA/FASAH public guidance and industry sources (see References)

Figure 1. Saudi import clearance process map, showing the primary error zone and the two downstream paths (fast release vs. cost cascade).

5.2 Scale of the Problem — Error Rate Benchmarks

While Saudi-specific, shipment-level error statistics are not publicly published, international benchmarking gives a reasonable proxy for scale:

15% of all customs declarations globally contain some documentation error (ICC, 2023)	23% of cross-border e-commerce shipments are delayed specifically by HS misclassification	SAR 5,000– 50,000+ typical penalty range for misdeclaration in Saudi Arabia, by severity
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5.3 Three Real-World Saudi Case Examples

To test whether the process-mapping findings generalize across product categories, three independently reported Saudi clearance incidents were reviewed:

- Case A — Riyadh industrial solvent importer: a chemical shipment was filed under an incorrect HS code, resulting in SAR 45,000 in fines and a two-week clearance delay. This illustrates a pure classification error on a high-scrutiny (chemical) product category.
- Case B — European electronics manufacturer, King Abdulaziz Port (Dammam): a shipment arrived and was held after customs flagged missing SASO conformity certificates; demurrage accrued while the importer scrambled to resolve the gap, and the underlying distribution deal nearly collapsed. This illustrates a missing-regulatory-approval failure, distinct from HS code error, with a direct commercial (not just financial) consequence.
- Case C — Textile/apparel importer: Saudi rules require correct HS classification alongside Arabic-language labeling (fabric composition, origin, care instructions) and an authenticated certificate of origin; a mismatch in any one of these three elements can trigger detention or market withdrawal. This illustrates a compounded-documentation failure, where the point of risk is a bundle of requirements rather than a single code.

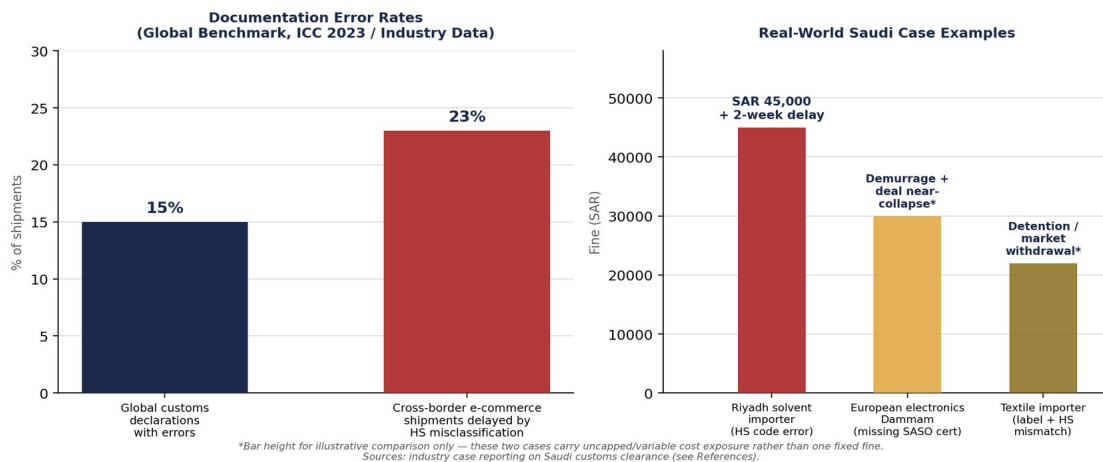


Figure 2. Left — global documentation error-rate benchmarks. Right — relative cost/consequence severity across the three reviewed Saudi case examples.

5.4 The Accountability Gap

The process mapping and case review converge on a single structural explanation for why these errors recur despite a modernized, digital-first customs system: the party who describes the product (the shipper), the party who

classifies it (the broker), and the party who bears the financial and legal consequence of an error (the importer) are three separate actors, none of whom owns the process end-to-end.

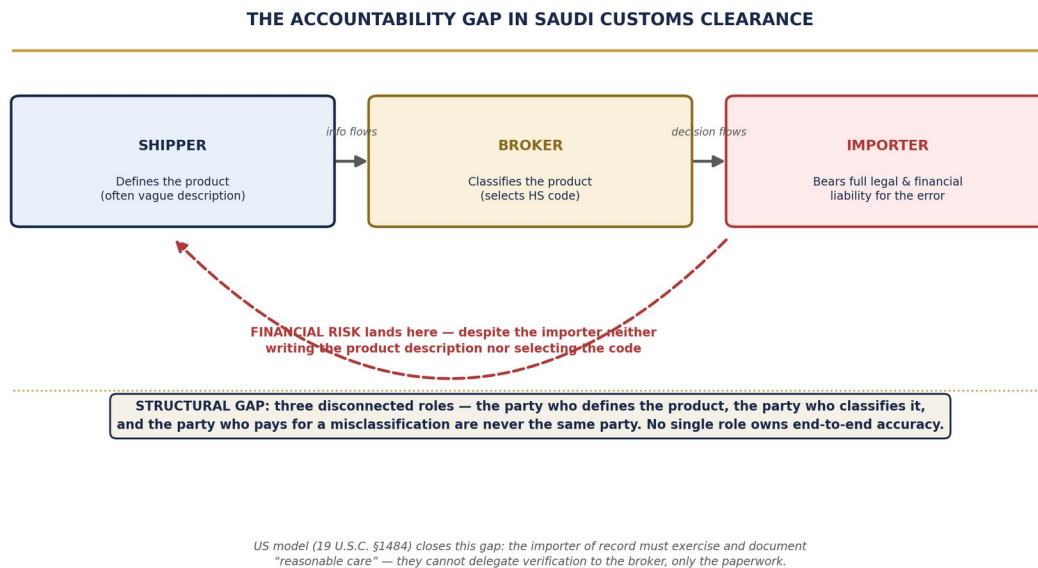


Figure 3. The accountability gap — decision-making and financial risk sit with different parties in the standard Saudi clearance workflow.

6. Discussion

6.1 Why a “Simple” Fix Is Rarely Implemented

A recurring pattern across the case examples is that the fix is not technologically difficult. Standardized pre-shipment documentation, a supplier-provided technical data sheet, or a second-look broker audit are all low-cost, low-complexity interventions. What is difficult is sustaining them as a daily discipline across multiple independent organizations (exporter, broker, importer) who do not share a single incentive structure. Companies tend to default toward purchasing more technology — a bigger ERP module, an automated classification tool — because it is a visible, one-time decision, whereas closing the accountability gap requires an ongoing, cross-organizational habit that no single actor is fully incentivized to enforce.

6.2 A Useful Comparison: The United States Model

The United States' import compliance framework offers a useful contrast. Under 19 U.S.C. § 1484, CBP holds the importer of record legally responsible for exercising and documenting “reasonable care” in classification, valuation, and origin determination — a business may use a broker, but cannot delegate the underlying responsibility to them. In practice, this is reinforced through supplier-level requirements (a technical data sheet and supplier-recommended HS code with justification), documented classification rationale retained for audit purposes, and active, periodic broker audits by the importer.

This does not remove the possibility of error — US importers still face significant classification penalties — but it structurally discourages the passive “importer trusts broker, broker trusts shipper” chain that this analysis identified

as the core failure mode in the Saudi cases reviewed. The principle, rather than the specific US legal mechanism, is the transferable insight: verification needs an owner.

6.3 Relevance to Entry-Level Logistics Roles

An important, and encouraging, finding for a candidate entering the field is that this is not a director-level problem. A pre-arrival documentation checklist, a supplier technical-spec template, or a scheduled broker-declaration audit are precisely the kind of process controls a Logistics Coordinator or Warehouse/Inventory Assistant could own and operate — they require attention to detail and consistency, not senior authority. This mirrors the FEFO and cold-chain logging discipline applied during hospital pharmacy inventory work: the pattern (small, repeatable, verifiable checks) is the same, only the product category and regulator differ.

7. Conclusion & Recommendations

This analysis set out to identify where, structurally, documentation errors originate in Saudi Arabia's import clearance process and why they persist despite significant digital modernization. The finding is that the error point is rarely the technology — FASAH, SABER, and ZATCA's risk-based inspection model are genuinely modern systems — but the accountability gap between the party who describes a product, the party who classifies it, and the party who pays for a mistake.

Recommended intervention (two-layer, low-cost):

- Layer 1 — Shipper-side: a standardized pre-shipment product specification template (material composition, technical use, prior HS code with justification) required before booking, closing the “vague description” failure mode identified in Section 5.3.
- Layer 2 — Broker-side: a mandatory second-look classification check against ZATCA's tariff database before Bayan submission, with the rationale documented and retained — adapting the US “reasonable-care” documentation practice to the Saudi workflow.

Projected impact (illustrative): avoiding a single 2-day clearance delay on one container saves an estimated \$300–600 in demurrage at published 2026 industry rates. Scaled across a modest monthly volume of 20 containers, consistent application of this two-layer check could plausibly avoid \$6,000–\$12,000 per month in preventable demurrage exposure for a mid-sized importer — before accounting for avoided fines, storage costs, or preserved business relationships of the kind nearly lost in the Dammam electronics case.

More broadly, closing this gap contributes to the operational reliability that underpins Saudi Arabia's Vision 2030 logistics and non-oil diversification goals — efficient, predictable customs clearance is a precondition for the FMCG, retail, and industrial growth the Kingdom is actively pursuing.

7.1 Illustrative Delay-Reduction Estimate (Supplementary)

This section was added to give a transparent, sourced answer to a question the original analysis did not directly address: roughly how much delay or cost reduction the proposed two-layer intervention could realistically achieve. No Saudi-specific, shipment-level before/after dataset is publicly available (see Section 4, Limitations), so no single validated percentage can be claimed as a measured outcome. The estimate below is instead constructed directly from the benchmarks already cited in Section 5.2, and should be read as directional, not empirical.

Basis for the estimate

Global benchmarking cited in this study indicates that approximately 15% of customs declarations contain some documentation error (ICC, 2023), and that HS misclassification specifically is linked to roughly 23% of cross-border e-commerce shipment delays. The two-layer control proposed in Section 7 (a standardized pre-shipment product specification, plus a second-look broker classification check) is designed to target exactly this failure mode — a vague product description or an unverified HS code — rather than delays caused by unrelated factors such as port congestion or vessel scheduling.

Directional estimate

If the intervention captures a meaningful share of the documentation-driven errors within that reported 15–23% range, a mid-sized importer could reasonably expect a low-double-digit percentage reduction in *preventable, documentation-related* clearance delays specifically — consistent in scale with the \$6,000–\$12,000/month illustrative savings estimated in Section 7 for a 20-container/month import volume. This does not imply a reduction in total port delay of that magnitude, since demurrage also arises from causes outside documentation accuracy (e.g., congestion, physical inspection backlogs, carrier scheduling).

Caveat: This figure is a directional estimate derived from published global benchmarks, not a measured Saudi outcome. It should be presented publicly with that context attached, consistent with the Limitations noted in Section 4.

8. References

Sources consulted during this analysis (publicly available industry, regulatory, and news publications, accessed August 2026):

- Zakat, Tax and Customs Authority (ZATCA) — public guidance on FASAH, Bayan filing, and the Integrated Customs Tariff.
 - Radhi Awad Customs Consultancy — “Customs Clearance in Saudi Arabia: Beginner's Handbook” and “Saudi Customs Clearance: Key Steps” (2025).
 - Bahkaly Logistics — “Customs Clearance in Saudi Arabia | Complete Guide” (2026).
 - Palm Horizon KSA — “Apparel & Textiles HS Codes in Saudi Arabia” and “Saudi Arabia's Import Rules Are Shocking International Businesses” (2026).
 - Fahes / Tabseer — “HS Code Saudi Arabia: Structure, ZATCA & SABER” and “HS Code Search Saudi Arabia” guides.
 - Kisun Shipping — “Saudi Restricted Imports: SFDA, GACA & CST Approvals” (2026).
 - International Chamber of Commerce (ICC) — 2023 global customs declaration error-rate study, as cited in DutyPilot, “7 HS Code Errors E-commerce Makes & How to Fix Them.”
 - Gaia Dynamics — “CBP Tariff Classification 101”, “How Incorrect HS Codes Lead to Unexpected Tariff Charges”, and “What Is an HS Code” (2026).
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 - U.S. Customs and Border Protection (CBP) — Customs Broker FAQ and Harmonized Tariff Schedule guidance.
 - ISF Customs Broker — “Step-by-Step Customs Clearance in the United States” (2026).
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 - Trade.gov — “Saudi Arabia — Import Requirements and Documentation.”
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— End of Case Study —